# Week 2 Presentation

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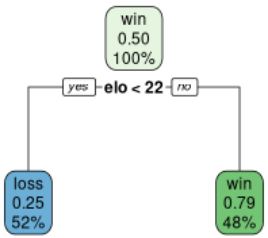
October 28, 2026

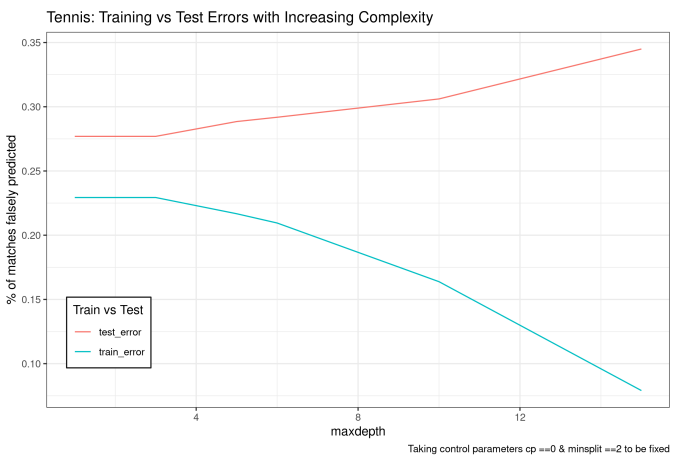
# Tennis Tree Regression

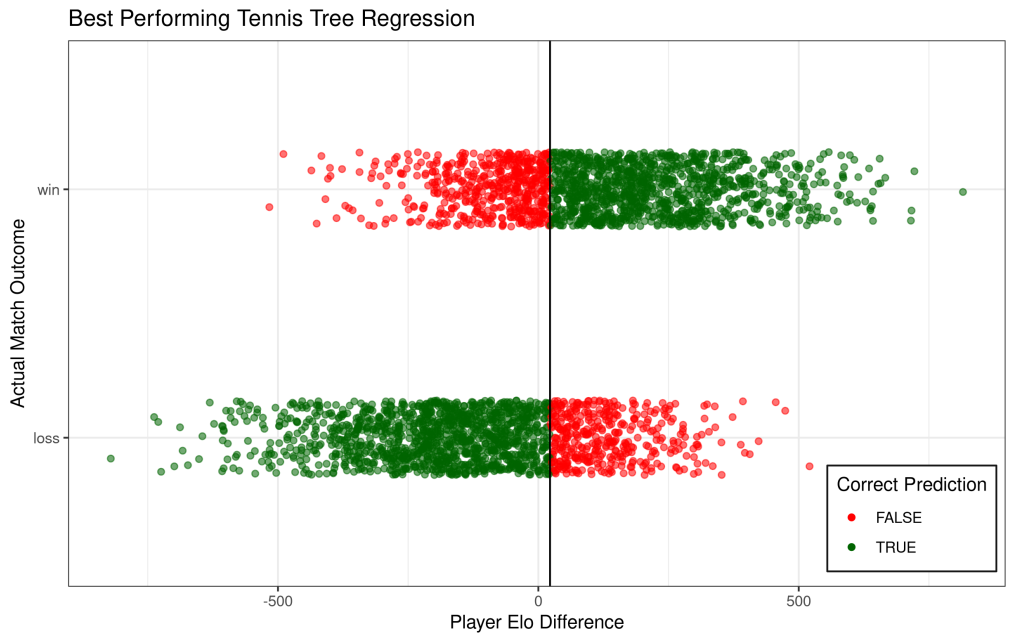
## Highlight of Tennis Grid Search Model Comparison

| comment | cp | minsplit | maxdepth | train_error | test_error | nsplits |
|---|---|---|---|---|---|---|
| least test error | 0.000 | 2 | 1 | 0.229 | 0.277 | 1 |
| reasonable performer using more than one split | 0.001 | 40 | 5 | 0.224 | 0.279 | 6 |
| worst overfit | 0.000 | 2 | 15 | 0.079 | 0.345 | 434 |

## Best model:



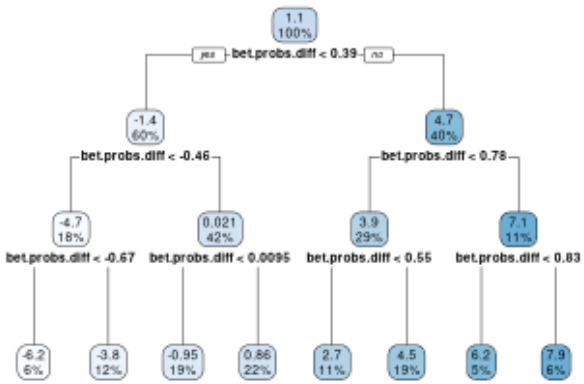



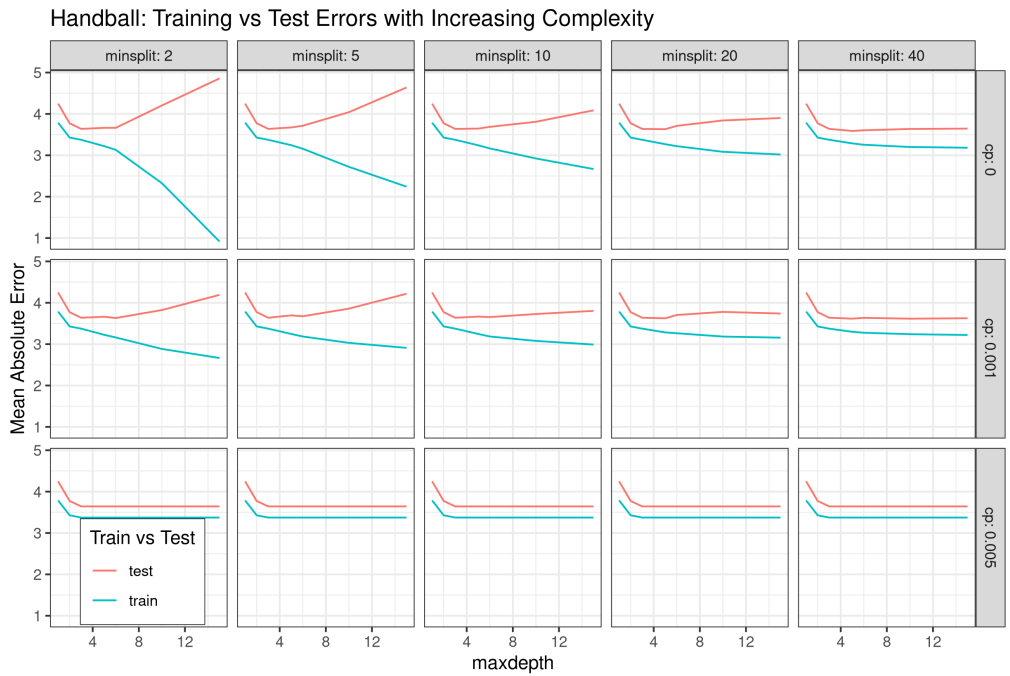
# Handball Tree Regression

## Highlight of Handball Grid Search Model Comparison

| comment | cp | minsplit | maxdepth | train_mae | train_rmse | test_mae | test_rmse | nsplits |
|---|---|---|---|---|---|---|---|---|
| best test mae | 0 | 40 | 5 | 3.288 | 4.213 | 3.588 | 4.582 | 25 |
| compromise btw number of splits & test mae | 0 | 2 | 3 | 3.375 | 4.307 | 3.638 | 4.606 | 7 |
| worst overfit in mae & rmse | 0 | 2 | 15 | 0.916 | 1.915 | 4.859 | 6.387 | 914 |

## Best compromise:



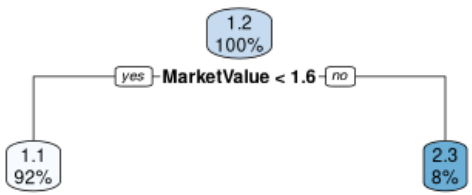



# Football Tree Regression

## Highlight of Football Grid Search Model Comparison

| comment | cp | minsplit | maxdepth | train_mae | train_rmse | test_mae | test_rmse | nsplits |
|---|---|---|---|---|---|---|---|---|
| best test mae | 0 | 5 | 3 | 0.780 | 0.980 | 0.741 | 0.985 | 7 |
| a single split model does well | 0 | 2 | 1 | 0.835 | 1.064 | 0.770 | 1.041 | 1 |
| worst overfit in mae & rmse | 0 | 2 | 15 | 0.112 | 0.312 | 1.326 | 1.716 | 171 |

## Best single split:




## Lowest MAE:

